THE USE OF HEDGES IN RESEARCH ABSTRACTS ACROSS APPLIED LIFE SCIENCES

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By

CHRYZ KEAN F. MANUEL
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INTRODUCTION

Background of the Study

Certainty, it seems, is one thing academic writers are certain they should avoid. Writing guides urge developing researchers to "*qualify your claims*," as if hedging were a single transferable skill. Yet the actual language of uncertainty shifts with the nature of the evidence and the demands of each field.

Expert writers hedge to present claims with caution rather than absolute certainty. Hyland's (1996) study of molecular biology research articles shows that hedging is not decorative caution. It is a core rhetorical resource that lets writers signal epistemic stance, mark membership in a scientific community, and state claims with precision while still showing deference to colleagues. Devices such as may, suggest, and generally recur in that expert prose because they manage how far a claim should be trusted. Hyland (1998) develops that account into a fuller model of hedges as means of reducing commitment to the truth of a proposition or of preferring not to state commitment categorically. Those expert norms in scientific English are well documented. Far less is known about undergraduate thesis writers, who face related genre expectations without the same length of disciplinary training. The open question is whether their abstracts already reflect field-specific conventions, or only a generic style of caution learned as a general writing rule.

That question matters in the applied life sciences. Research in Agriculture, Veterinary Medicine, and Biology can shape community practice

and environmental decisions. When developing researchers state provisional findings as settled fact, overstated claims can travel further than the evidence supports. Managing epistemic stance is therefore both a writing skill and a practical responsibility.

This study examines undergraduate thesis abstracts from three programs at Pampanga State Agricultural University (PSAU): Agriculture, Veterinary Medicine, and Biology. These fields share an institutional context but differ in how they typically handle evidence. Agricultural research often involves field conditions outside the researcher's control. Biological studies frequently take place under controlled laboratory conditions. Veterinary medicine manages clinical variables tied to individual patient outcomes. The boundaries are not absolute, since agricultural work can be lab-based and biology can be field-based, but the different epistemic orientations may still shape hedging. If disciplinary norms appear at the undergraduate level, they should show up here. If they do not, the shared pattern is equally informative for instruction.

Hedging in Undergraduate Writing

Studies of developing researchers converge on a restricted hedging repertoire, but the evidence is not a single undifferentiated claim. Khunasathitchai et al. (2025) analyzed move structure and hedge use in 60 research paper abstracts written by Thai EFL English-majored undergraduates. Three moves were obligatory in their sample: presenting the research, describing the methodology, and summarizing the findings. Hedges appeared in the moves that present the research, summarize findings, and discuss the study, not evenly across the abstract. Modal verbs were the most frequent hedge

type, followed by epistemic lexical verbs and epistemic adverbs, while epistemic adjectives were rare and epistemic nouns were absent. The inventory is therefore small and modal-heavy even in a language-major sample writing the high-stakes abstract genre. Suratno and Aydawati (2025) shift the lens to discussion sections of non-science undergraduate theses. Using a corpus of 30 theses (over 100,000 words), they found that student writers relied heavily on the modal dapat compared with other hedging devices. That concentration suggests limited familiarity with a wider hedge set. Because discussion sections offer more rhetorical space than abstracts, the finding is telling: restricted variety is not only a product of length pressure. It also reflects a developing stance repertoire that has not yet branched into the fuller range of forms common in expert writing.

Ghia et al. (2022) supply one of the clearest abstract-focused accounts. They built a corpus of 217 English-language bachelor's thesis abstracts written by advanced Italian EFL undergraduates specializing in linguistic and cross-cultural mediation, totaling 287,048 words, and compared hedge use with a subsection of successful L1 student writing in MICUSP. Fifteen hedge forms were set against boosters and modals of logical possibility and inferential certainty. The EFL abstracts used fewer hedges overall and leaned more on boosters and on modals of epistemic certainty than the L1 comparison group. At the same time, patterns in the two corpora still showed parallelisms, which the authors read as awareness of hedging strategies rather than absence of the concept. The study matters for the present work because it isolates the thesis abstract as a genre and shows that advanced student writers can under-hedge even when their programs center on language. Mazidah et al. (2022) compared

hedges and boosters in 50 undergraduate thesis abstracts from Qomaruddin University: 25 from an English Language Education program and 25 from a Mathematics program. Coding followed Hyland's (2005) taxonomy throughout. Across the sample they recorded 53 hedge tokens (22 variants) against 130 booster tokens (17 variants), so boosters outnumbered hedges overall. Mathematics abstracts used more hedge tokens than the English-education abstracts, while the English-education group still showed a wider set of hedge word variants. The authors link student hedging to limited knowledge of English academic rhetoric. For the present study, the result underscores that undergraduate abstracts can favor certainty markers as much as caution, and that program differences need not follow a simple language-versus-science split.

Wang (2022) widens the student-writing picture by comparing hedging across sixteen disciplines in MICUSP. Five types of hedging devices were examined in successful upper-level student papers rather than in expert research articles alone. Hedging was more common in argumentation-driven disciplines than in data-driven ones, and further differences appeared between disciplines housed under the same broad division. That pattern challenges a simple natural-versus-social science split: two science fields need not hedge alike, and two fields within the same division need not either. Modal auxiliaries were one of the device types tracked, and disciplinary profiles for modal-auxiliary hedges differed across the corpus. Wang's study therefore supports disciplinary comparison among student writers while also showing that aggregate "more hedging" can hide different formal mixes. For applied life-

science undergraduates, the implication is that any cross-program difference at PSAU should be read against this dual finding: discipline can matter, and form-level patterns need to be reported alongside density.

Halabisaz et al. (2014) focus the comparison on thesis abstracts in applied linguistics. They sampled 300 master's abstracts written between 2000 and 2013, split between English L1 writers and Iranian student writers, and coded hedges according to Crompton's (1997) taxonomy. English L1 writers used more hedging devices overall, and student writers used fewer. The authors attribute the difference to rhetorical sensitivity, modality, awareness of audience and purpose, and cultural background. Because the unit of analysis is the thesis abstract, the study sits closer to the present design than full-paper comparisons do. It also shows that L1-second-language differences in hedging can appear even among writers who have completed substantial graduate coursework in a language-related field.

Taken together, these studies show that undergraduate and master's-level writers often under-deploy hedges, lean on a limited set of forms (often modals), and may differ by program without matching expert disciplinary profiles. Clear gaps remain for the present work: only a few of these studies examined abstracts specifically, and most of those focused on English or language majors rather than applied science programs. Others analyzed discussion sections or full papers, where rhetorical space is wider. None is set in a Philippine context for Agriculture, Veterinary Medicine, and Biology undergraduate thesis abstracts.

Disciplinary Variation in Scientific and Medical Writing

Field norms shape expert hedging, but not along a simple hard-or-soft divide. Hyland (1996) grounds that claim in scientific research writing itself. In a corpus of 26 molecular biology research articles, he shows that hedging is central to gaining reader acceptance of claims. Writers use hedges to state findings with precision, caution, and deference to colleagues rather than as optional politeness. Lexical forms carry most of the load: about 79% of hedging instances are lexical, mainly verbs, adjectives, and adverbs, rather than modal verbs alone. That formal profile matters for later comparison with undergraduate biology and applied-life-science abstracts that may lean more heavily on a small modal set.

Afshar et al. (2014) scale the comparison across fields. They examined Discussion sections of 180 research articles from Humanities (Law, Persian literature, TEFL), Basic Sciences (Biology, Applied Chemistry, Geology), and Agriculture (Biotechnology, Soil Sciences, Horticulture), with 20 articles per sub-field, using Salager-Meyer's (1994) taxonomy. Significant differences appeared both among the three fields and among sub-fields within each field. Humanities articles carried the highest overall hedge counts. Agriculture and Basic Sciences sat lower, with further unevenness inside each label. The study therefore does more than rank soft above hard. It shows that sub-field matters even when articles share a broad disciplinary name.

Mohammed (2011) takes a different cut by comparing lexical hedges in biochemistry and applied-linguistics research articles by American and Arabic authors. The study found that American writers used more hedging forms than Arabic writers in both disciplines. Linguists used considerably fewer hedges than scientists, and biochemistry articles were more heavily hedged than

applied-linguistics articles. Discussion sections contained more hedges than introductions. Because Mohammed focuses on lexical hedges rather than Salager-Meyer's full taxonomy, the ranking cannot be stacked directly onto Afshar et al.'s field totals. What the two studies share is that density tracks the pairing under analysis and the coding scheme, not a single universal hard-or-soft scale.

Rezvani and Javadi (2021) test hard-versus-soft differences in abstracts by Iranian writers of English. They sampled 200 research-article abstracts, 50 each from chemistry, mathematics, philosophy, and English, grouped as hard and soft sciences under Biglan's (1973) typology. Soft-science abstracts used more hedge expressions overall, and soft writers drew more on modals, verbs, nouns, adjectives, and adverbs, while hard-science writers used more conditional expressions. Epistemic and cognition verbs were the most frequent hedging expressions in both groups. The study supports disciplinary differentiation inside the compressed abstract genre and shows that form mix, not only density, separates the two clusters.

Bonyadi, Gholami, and Nasiri (2012) keep the lens on an applied science close to agriculture. They compared hedges in Discussion sections of 60 environmental-sciences research articles: 20 by English research writers, 20 by Iranian writers in English, and 20 by Iranian writers in Farsi, coded with Salager-Meyer's (1994) taxonomy. English and Iranian English articles did not differ significantly in hedge use, but English and Farsi articles did. The authors link the English-Farsi gap to rhetorical preference in Farsi for less hedged, more categorically validated prose, a pattern that sits at odds with international environmental-science discourse expectations. For the present corpus, the

study is a reminder that applied-science hedging norms are international as well as disciplinary, and that L1 background can still shape English research writing in the same field.

Within life sciences proper, Souici and Slougui (2022) examined 31 biology research articles by Algerian authors from five locally published journals. Main verbs were the most prominent hedge category. Within that category, non-factive reporting verbs ranked highest. Writers relied on those verbs to support and frame findings rather than on a wide modal set alone. Hamidi et al. (2025) compared conclusion sections of basic-science and clinical-science papers in the Medical Journal of the Islamic Republic of Iran. Clinical papers used more hedge tokens (330 versus 154), yet an inferential test found no statistically significant difference. Journal standards and cultural factors, the authors argue, shape hedging as much as the basic-versus-clinical split.

Salager-Meyer (1994) anchors much of this comparative work in medical English. In research papers and case reports from leading medical journals, hedge frequency and category distribution shifted with rhetorical section. Discussion and Comment sections were the most heavily hedged. Methods and case-report body sections were among the least. Shields, approximators, and compound hedges accounted for the great majority of devices. Introductions favored shields, especially epistemic verbs, as writers carved an early niche for the study. That section pattern matters for how medical writers present results. Directly measured findings often remain categorical, and hedging rises where writers move from measurement to interpretation, generalization, or clinical implication. The same medical

discourse can therefore look firm in the results strip and still manage claim responsibility in the interpretive strip.

Corpus work on medical Results and Discussion sections supports that reading. Doğan and Akbaş (2021) analysed about 215,000 words from the Results and Discussion sections of 100 medical research articles across fields from anatomy to endocrinology. Medical writers used modal auxiliaries frequently to mark stance. They did not rely on a matching surge of boosters to amplify commitment. Where boosters appeared, they were often strategic, used to support credibility and reader engagement rather than as a default claim of certainty. Medical scientific English in those sections is therefore better described as calibrated commitment than as either blanket tentativeness or blanket conviction.

Lingard (2026) makes the same point in pedagogical terms for health-professions research writing. High authorial commitment is appropriate when the warrant is strong. Heavy hedging on a well-supported measurement can read as unnecessary softening. Low commitment remains necessary when the claim outruns the data. The issue is match of stance to warrant, not a medical culture of always agreeing or disagreeing more bluntly than other sciences.

Cross-field comparisons complicate any simple medical-versus-other ranking, as Sepehri, Hajijalili, and Namaziandost (2019) compared hedges and boosters in 30 medical and 30 engineering research articles and found a significant disciplinary difference, with medical articles carrying more hedge tokens overall in their sample. Medical writing is not uniformly under-hedged relative to other technical fields. What remains distinctive is the reporting pattern inside the article: categorical presentation of measured results, denser

hedging where interpretation and generalization begin. Hu and Cao (2011) add publication medium and evidence type outside medicine. In applied-linguistics journal abstracts, English-medium texts featured markedly more hedges than Chinese-medium texts, and empirical research abstracts used significantly more boosters than non-empirical articles. Stance choices therefore track section, medium, and evidence type as well as discipline.

Smirnova and Pérez-Guerra (2025) ask whether formal stance features alone can recover hard-versus-soft disciplinary groups. Their corpus covers about 1.6 million words of research articles from eight disciplines: chemistry, physics, engineering, and mathematics on the hard side, and business studies, history, linguistics, and political science on the soft side. Grammatical and lexical stance features, including hedges, boosters, and self-mention, were quantified and modelled. Multivariate analysis identified features that significantly contributed to hard-versus-soft variation. Clustering on those formal frequencies alone grouped the eight disciplines into two clusters matching the original hard and soft division. The result strengthens the claim that disciplinary variation in stance is systematic enough to classify fields, even when meaning-based interpretation is set aside.

Expert scientific English therefore shows measurable variation, but the variation is local to how each field, section, medium, and publication context handles evidence and claim risk. In medical and clinical writing, that local pattern includes categorical reporting of measured results and denser hedging when claims move into interpretation. For developing writers in second-language settings, the expert picture is a target rather than a description of current practice. The present study asks whether PSAU undergraduates in three

applied programs already show any of that field-sensitive differentiation in the compressed genre of the thesis abstract.

Hedging in L2 and EFL Contexts

English is the medium of academic writing at PSAU, but it is not the home language for most students. Research on second-language academic writing has long treated hedging as a point of difficulty rather than as a simple vocabulary gap. Hyland and Milton (1997) compared expressions of doubt and certainty in a one-million-word corpus of examination scripts from 900 Cantonese-speaking school leavers writing in English and 770 British learners of similar age and educational level. The L2 writers relied on a more limited range of epistemic items, offered stronger commitments, and showed greater difficulty conveying a precise degree of certainty than their L1 peers. That pattern remains a baseline across EFL research. L2 academic writers often under-qualify claims or overuse a small set of strong forms.

Hinkel (2005) sharpens the formal picture with a large student-essay corpus of 745 essays and about 221,000 words. L2 writers used a limited range of hedging devices, many of them associated with conversational rather than formal written discourse. The same essays showed a prevalence of conversational intensifiers and overstatements that are common in informal speech but rare in academic prose. Demir (2018) compared lexical hedges in academic texts by English L1 and L2 writers and reported significant differences across modal, verbal, adjectival, adverbial, quantifier, and noun hedges, as well as in total hedge use. Taken together, these studies show that the L2 issue is not only density. It is range, register, and the balance between caution and overstatement.

Takimoto (2015) adds a recognition layer. In a corpus comparison of academic essays by Japanese EFL learners and English L1 writers, the EFL group used more boosters than hedges and preferred straightforward, explicit expressions, whereas English L1 writers were more tentative. A follow-up task suggested that certainty items were more visible to the learners than possibility and probability items, which they often misapprehended. Underuse of hedges can therefore coexist with limited pragmatic awareness of what the missing forms do. Luo (2021), in a systematic review of undergraduate English language learners' hedging, reaches a similar aggregate conclusion. ELLs generally use fewer hedges than English L1 writers, draw on a narrower inventory, and face combined challenges of academic experience and L2 form control.

Alonso (2019) approaches the same problem through a multicompetence design rather than an L2-only comparison. She examined hedges in academic writing by 24 Spanish learners of English, collecting data from the same participants in both Spanish and English, and compared them with monolingual Spanish and monolingual English control groups of 24 speakers each. Hedges were coded following Hyland's (1994) typology. Chi-square analyses showed differences between English and Spanish speakers, and between English speakers and L2 learners, especially in impersonal expressions and lexical verbs. Evidence of crosslinguistic influence appeared, and similarity in the use of adverbs across the learners' L1 and L2 pointed to L1-L2 convergence. The design matters for the present study because it shows that L2 hedging patterns cannot be read as a pure deficit against English L1 norms without also considering the writer's full language system. Publication medium and cultural writing tradition can widen the same gap. Hu and Cao (2011) found

that applied-linguistics abstracts in English-medium journals carried markedly more hedges than those in Chinese-medium journals, a difference they link jointly to rhetorical convention and epistemological belief.

Undergraduate thesis abstracts, the genre closest to the present corpus, show the same L2 pressures. As reviewed under undergraduate writing, Mazidah et al. (2022) and Ghia et al. (2022) report that L2 thesis abstracts often carry fewer hedges and stronger certainty marking than L1 comparison groups, even among language-oriented programs. Bayraktar-Cepni and Kulaksiz (2022) compared hedging in discussion sections of ELT MA and PhD theses and found that MA theses contained significantly more hedges than PhD theses, a difference they link to developing writers′ stronger need to soften stance. Level of academic experience, not only L1 versus L2 status, therefore shapes how much caution writers encode.

Instructional research points in the same direction. Petchkij (2019) argues that explicit teaching of hedges can raise their use in Thai EFL contexts, though modal forms often remain dominant. Loi and Lim (2015) analysed hedging orientations in discussion sections from the Journal of English for Academic Purposes and elicited specialist views on L2 difficulty. Those specialists attributed second-language learners′ difficulty modulating claims to socio-cultural factors, classroom instruction, disciplinary culture, and disciplinary appeals, and stressed that pragmatic fluency in hedging is teachable rather than automatic. Regional ASEAN and Philippine learner studies, taken up in the next section, likewise report limited hedge inventories and frequent booster preference in student academic English (Ningrum et al., 2024; Lopez et al., 2025; Escalona, 2025). Since the present study does not

measure English proficiency independently, observed patterns in the PSAU corpus may reflect L2 constraints, disciplinary socialization, instructional history, or some combination of these. The aim is a descriptive baseline of how stance is worded in ratified undergraduate thesis abstracts, not a test of which single factor causes the pattern.

Hedging in Philippine Academic Writing Contexts

Philippine and regional findings align with the broader picture of English-medium student writing while adding local detail. Ningrum et al. (2024) analysed academic writing from the ICNALE learner corpus across ten ASEAN contexts, including the Philippines. ASEAN EFL learners preferred boosters over hedges. Overused boosters included completely, really, and always, while the most frequent hedges included usually, often, and maybe. The pattern suggests that regional learners often encode confidence more readily than caution in academic prose.

Local college essay studies point in the same direction. Lopez et al. (2025) examined 50 argumentative essays by English Language Studies majors at a northern Philippine state university. Students showed developing-stage control of hedges and boosters, with epistemic modal verbs the most frequent forms across proficiency levels. Escalona (2025) compared hedges and boosters in essays by AB-English and BS-Information Technology students and found that female writers used hedges more often than male writers, while overall frequencies remained relatively consistent across major fields. Both studies examine essays rather than thesis abstracts, so they cannot be read as direct predictions for the present corpus. They do show that Philippine undergraduate

academic English often leans on a small modal set and mixes intensifying moves with limited softening.

Research on Philippine student research writing brings the evidence closer to the thesis genre. Luyun (2024) analysed introduction and conclusion sections in 41 undergraduate research papers from BSED and BEED programs at a Philippine university, using Mojica's hedge categories and Hyland's booster taxonomy. The most frequent hedge forms included this study, it, and the study, with may also appearing among Type I devices. The dominant booster was should, and no significant difference in hedge or booster frequency emerged between the two education programs. Melon-Galvez (2017) examined the hedging system of Filipino engineering students writing technical ESP texts and linked hedge use to framing, intertextuality, and task type, with pedagogical implications for precision and interpersonal stance in technical discourse.

Published research articles written in Philippine English add a professional layer. Magday et al. (2026) compared hedges in the Results and Discussion sections of 70 English Language Teaching research articles, 35 by Filipino writers and 35 by Thai writers. Filipino writers used hedges at a higher density (23.74 per 1,000 words) than Thai writers (17.63 per 1,000 words). The authors link the difference in part to the Philippines' ESL relationship with English versus Thailand's EFL context, even where disciplinary conventions are shared. Gamad et al. (2024) analysed metadiscourse markers in the Results, Discussion, and Conclusion sections of 90 research articles across social sciences, linguistics, and business in Philippine English, American English, and Chinese English. Hedges were the most frequent interactional markers overall,

and Philippine English research articles showed comparatively high metadiscourse use, with interactional preferences varying by discipline.

Outside student essays and research articles, Philippine editorial writing shows hedging as a context-sensitive resource rather than a fixed national style. Villanueva (2014) examined 30 editorials from three leading Philippine newspapers and found that nearly half of the hedges were references to official reports or authorities, with modals and adverbs making up a little over 30 percent. Those forms helped editors manage accountability and soften claims on politically sensitive topics. Grengia (2025) extended the editorial analysis to student publications, The Philippine Collegian and The LaSallian, and found deliberate but publication-specific hedge use tied to voice, audience, and rhetorical goals. Taken together, the Philippine evidence shows modal-heavy inventories, frequent booster preference in learner writing, and higher hedge density in some professional Filipino ELT research than in comparable EFL contexts. No published study identified in this review has examined hedging in undergraduate thesis abstracts from applied life sciences in a Philippine setting. That gap is the present study's descriptive starting point.

Disciplinary Variation in Abstracts and Undergraduate Theses

Abstracts compress research into a constrained space. Expert abstract studies already show that field norms reach this short genre. Imataki and Kobayashi (2024) report measurable disciplinary differences in expert abstracts for features such as voice and nominalization. Xie and Mi (2023) examine promotion and caution across disciplines and ranking levels. Hedge words were more frequent in soft sciences and higher-ranking journals, while positive promotional wording was more common in hard sciences. Their

results make abstracts a useful site for studying how caution and promotion compete under length pressure.

Thesis and dissertation abstracts written by developing researchers add a second layer. Wahyuningsih and Sofwan (2014) analyzed hedges in graduate thesis abstracts at Semarang State University across five programs: English, Mathematics, Science, Social, and Education Management. Lexical hedges were coded with Salager-Meyer's (1994) and Hyland's (1998) taxonomies, and non-lexical forms followed Navratilova (2013). English, Social, and Education Management abstracts used more hedges than Mathematics and Science. English led the lexical counts overall. Agentless passives were the dominant non-lexical form in every program. The authors link higher soft-field density to interpretive claims and lower hard-field density to measurement-based reporting. The study shows that disciplinary contrast appears inside the thesis abstract even when writers share one institution and one English medium.

Demanou and Tabe (2022) examined hedging in DIPES II dissertation abstracts at the Higher Teachers' Training College of Maroua across Bilingual Letters, English, Geography, and History. A specialized corpus of 46,368 tokens was searched with AntConc. Overall hedge density was low, about 2.6% of tokens. Markers of intentional vagueness, accuracy hedges, and writer-oriented hedges were the most common types. Students relied mainly on simpler forms such as some, few, many, could, and a number of. More complex frames such as it appears that were almost absent. Bilingual Letters abstracts were the most tentative, followed by English, Geography, and History. Across communicative moves, the findings and results move carried the densest hedging, especially writer-oriented forms that distance the writer from the

claim. The study treats hedging as both discipline-specific and move-specific inside the abstract.

Al-Zarieni and Alkhresheh (2024) studied hedges and boosters in 30 linguistics M.A. thesis abstracts at Yarmouk University, following Hyland's (2005) interpersonal model. Hedges outnumbered boosters (117 to 60, or 66% to 34% of the combined set). Hedges were read as ways of limiting opposition risk, marking politeness, and presenting results cautiously. Boosters were used to assert originality and stress certainty. The study adds an Arabic postgraduate linguistics sample to the abstract-focused record and shows that even language-focused programs still balance caution against certainty marking under abstract constraints.

Mazidah et al. (2022), reviewed under undergraduate writing, found a soft-versus-hard contrast in undergraduate thesis abstracts at Qomaruddin University that aligns with the graduate pattern reported by Wahyuningsih and Sofwan (2014). English-education abstracts used more hedges, while mathematics abstracts used more boosters.

Chafidha and Mujiyanto (2021) compared hedges in thirty master's thesis abstracts from the English and Indonesian graduate programs at Universitas Negeri Semarang, coded with Hyland's (1998) polypragmatic model. Writer-oriented, accuracy-oriented, and reader-oriented hedges appeared in both programs. English abstracts used more writer-oriented hedges, which distance the writer from the proposition. Indonesian abstracts used more reader-oriented hedges, which engage the audience. The study shows that program language focus can shape interpersonal orientation even when hedge functions themselves look similar.

Expert abstract conventions are therefore relatively well mapped for selected linguistic features and for caution versus promotion. Developing-researcher thesis abstracts show a recurring soft-versus-hard density contrast, a preference for simpler hedge forms, and program-level differences in interpersonal orientation. Most of that work still centers on language, education, or social-science programs. Whether undergraduates in applied life sciences show any disciplinary differentiation in this genre remains an open empirical question.

What the review establishes is a set of durable patterns rather than a single ranking of fields. Expert writers adjust hedge density and form mix to the epistemic demands of their work, and that adjustment is visible even inside the short abstract. Density alone does not tell the full story. Form mix, interpersonal orientation, and the move in which a hedge appears also vary by field and by communicative purpose. Developing researchers, including advanced student writers, typically work with a smaller set of forms and lean on modals. That restriction holds not only in abstracts, where length pressure is high, but also in longer sections such as discussions, which suggests a developing stance repertoire rather than a pure genre effect. Regional student writing often pairs the modal preference with relatively strong boosting, so caution and certainty marking coexist in the same short texts rather than trading off cleanly.

Thesis and dissertation abstracts written by students add a further constraint. Soft-versus-hard density contrasts and program-level differences in interpersonal orientation appear inside that genre, yet almost all of that evidence comes from language, education, or social-science programs. Applied

life sciences sit in a different relation to evidence. Claims are often tied to experimental outcomes, clinical observation, or field measurement, and the abstract must still package those claims under strict length limits. The open question is not whether undergraduates hedge at all. It is whether disciplinary differentiation of the kind documented for experts and for student writers in other fields appears inside applied life-science thesis abstracts under Philippine institutional conditions, and whether that differentiation shows up in density, form mix, function, or some combination of the three.

That empirical question cannot be answered without a clear boundary on what counts as a hedge. The literature does not share one definition, and the differences matter for coding. Hyland's polypragmatic model keeps both reduced commitment to propositional truth and reader-oriented interpersonal effects inside the same broad category. A single token can manage accuracy and manage the reader at once. Salager-Meyer treats the same devices as ways of distributing claim responsibility among writer, data, and circumstance, which is especially relevant for research writing where findings are provisional. Crompton argues for a narrower scope limited to epistemic modality on knowledge claims, on the ground that stretching hedge to every politeness feature of academic prose weakens the term and blurs what is being counted.

Directives and recommendations sit at the edge of that debate. They manage action rather than the truth of a claim. Under a wide interpersonal reading they look like mitigation. Under a claim-centered reading they fall outside the hedge count. This study adopts the claim-centered boundary, so a token is retained as a hedge only when it changes the writer's stance toward a proposition that evidence could show to be wrong. Epistemic Caution records

reduced commitment to that claim's truth. Politeness/Deference records interpersonal softening of the same claim when that softening co-occurs with, or stands in place of, an epistemic effect on a claim. Softened recommendations are logged as false positives under this scope. The choice is a position in the definitional debate.

The empirical gap was checked through searches of Google Scholar, ERIC, and ScienceDirect (2010–2025) using combinations of hedging, hedges, epistemic stance, undergraduate, thesis abstract, research abstract, Agriculture, Veterinary Medicine, Biology, Philippines, and second language, with follow-up citation chaining from key reviews and regional studies cited above. No published study identified in that search has examined hedging in undergraduate thesis abstracts from applied science disciplines in a Philippine context.

The present study takes up that intersection at PSAU. It maps hedge type, density, and function across three applied life sciences programs that share one institution and one English medium but differ in how closely their claims are tied to experimental and clinical evidence. Agriculture, Veterinary Medicine, and Biology are treated as a local comparison set rather than as proxies for a global hard-or-soft divide. The aim is descriptive and comparative within that set: to show what hedge patterns appear in ratified undergraduate abstracts in these programs, and to read those patterns against the definitional and disciplinary background laid out above.

Statement of the Problem

Academic writing in the applied life sciences requires discipline-sensitive management of uncertainty in English. For English Language Studies,

that problem is about how student writers use the language's resources for stance when they compress research into an abstract. Prior work shows that expert hedging varies across fields such as agriculture and the basic sciences, yet there is no local corpus baseline for whether developing researchers at PSAU reflect those distinctions in their English abstracts, or whether they rely on a generic, undifferentiated approach.

This study mapped hedging patterns in undergraduate thesis abstracts from Agriculture, Veterinary Medicine, and Biology at PSAU. Specifically, it asked the following questions:

1. What types of hedging devices appear in the undergraduate thesis abstracts across the three disciplines?

2. How does the frequency of hedging devices vary across Agriculture, Veterinary Medicine, and Biology programs?

3. Among the hedging devices identified, what rhetorical functions do they serve?

4. What do the frequency and functional patterns suggest about the construction of epistemic stance through hedging in the ratified abstracts of developing researchers at PSAU?

Research Framework

This study is grounded in a coding scheme adapted from Hyland's (1998) classification of hedging forms, with function coded through an epistemic Working Test and a Reader Test under a claim condition. Figure 1 presents the conceptual framework. The grouping variable is discipline (Agriculture, Veterinary Medicine, and Biology). The dependent measures are hedge frequency, formal type, and function. The figure also marks the boundary

between retained epistemic caution and false positives excluded under the decision rules.

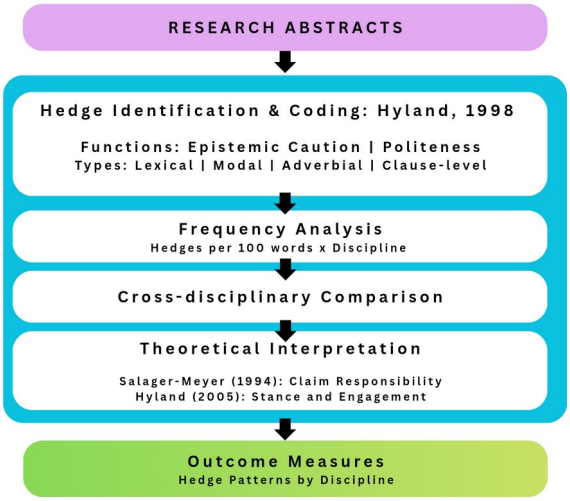
Figure 1

Conceptual Framework of the Study

Hyland (1998) is the type instrument and the broad start for epistemic caution and politeness/deference. Crompton (1997) supplies the boundary rule for epistemic caution alone: his restatement test operationalizes the Working Test for epistemic caution at Step 2. The study does not adopt Crompton's full taxonomy. Salager-Meyer (1994) is an interpretive lens in Part IV. Her claim-responsibility account helps read density differences across programs. Hyland (2005) is a second interpretive lens. The stance model helps read what fewer hedges mean where results turn into implications. Neither interpretive lens is an operational coding criterion.

Coding followed the decision rules in Appendix A. After a holistic reading of each abstract, Step 1 marked all possible hedge candidates without counting retained hedges. Step 2 applied a claim condition and two removal checks. Because a hedge changes the writer's stance toward a proposition, the device must attach to a claim: a statement about data, the world, or prior work that evidence could show to be wrong. Aims, directives, and routine procedures are excluded by kind as a consequence of that definition, not as a coding convenience. Under the Working Test, the device is removed from the claim. If the writer's commitment to the truth of the claim rises, the token is epistemic caution. Under the Reader Test, the device is removed from the claim. If commitment stays the same but the claim reaches the reader more bluntly, or the writer takes sole ownership against prior work, the token is

politeness/deference. If neither check is met, the token is a false positive. Retention and function assignment are the same operation by design. A token qualifies as a hedge by passing at least one test, and which test it passes is its function. Epistemic caution and politeness/deference may be assigned together when a token both reduces commitment to the claim's truth and softens how the claim meets the reader or prior work. In that case the Working Test still holds, and politeness/deference is recorded as a concurrent interpersonal reading rather than as a second independent pass of the Reader Test. The Reader Test alone is reserved for pure politeness/deference, where commitment does not rise. Step 3 assigned grammatical type only to retained epistemic caution and politeness/deference tokens. Residual rejects are labeled with reason codes such as recommendation, capability, procedure, or general false positive. Among retained hedges, the research questions on function ask how tokens split across epistemic caution and politeness/deference under this scope, not what functions hedges might serve outside it.



Significance of the Study

For writing instructors and thesis advisers at PSAU, the study provides an empirical baseline of how uncertainty is managed in the ratified abstracts of three applied life sciences programs. That baseline can support more targeted feedback than generic advice to "*be cautious*."

For academic writing researchers, it adds Philippine applied-science abstract data to a body of work that has drawn mainly from other regional and disciplinary settings.

For the BAELS program, it shows that corpus-assisted discourse analysis is a viable undergraduate thesis methodology and yields findings usable in local writing pedagogy.

For future researchers, the sampling design, coding decision rules, and false-positive log offer a replicable protocol for similar corpus studies.

Scope and Limitations

This study focused on undergraduate thesis abstracts from Agriculture, Veterinary Medicine, and Biology at PSAU. The inclusion window refers to the year of hard-bound submission as recorded in the PSAU library catalog, not the year the research was conducted. Theses conducted earlier but bound and catalogued in 2023-2025 are included. This reflects institutional practice in which submission date, particularly for irregular students, may postdate the research period by several years. The corpus consisted of 60 abstracts (20 per discipline, with Agriculture split evenly between Crop Science and Animal Science). Sampling design, stratum intervals, inclusion/exclusion criteria, and population and sample sizes are detailed under Research Corpus and in the Results chapter. Sub-programs are analyzed as a single "Agriculture" discipline

for cross-disciplinary comparison. No separate inferential comparison between them is undertaken.

For a manually coded functional analysis within a single institution, 60 texts are treated as sufficient for identifying patterns within the limits of a BA thesis. Findings are indicative patterns in the PSAU context. The pilot comparison appears in Appendix B.

The study is delimited to complete English-language abstracts. Abstracts with significant Filipino code-switching are excluded to maintain linguistic consistency. English proficiency is not measured, so the study makes no claims about learner English as such. Future work with proficiency controls could separate proficiency effects from disciplinary ones.

The abstract genre's rhetorical compression may suppress hedging variety compared to full-text discussions. Findings reflect hedging in this constrained genre only and should not be extrapolated to developing researchers' hedging competence in longer texts without further research. Additionally, this study is primarily delimited to hedging devices. A limited count of assertive reporting forms was performed only on the 25 zero-hedge abstracts. No comparison with hedged abstracts was made, and a full booster analysis was not undertaken. While credibility in academic writing involves a balance of caution and certainty, this study focuses specifically on the management of uncertainty.

A second coder was not available for inter-rater reliability testing, which limits how far the coding scheme can be verified against an independent application of the same decision rules. That limit bears especially on the functional findings in Part III. Assigning all 65 retained hedges to Epistemic

Caution may reflect one coder's frame as much as a property of the corpus. Inter-rater coding on at least a subset of the corpus is needed before those functional results can be treated as established. Coded data were reviewed with the thesis adviser for consistency with the decision rules in Appendix A.

The study also lacks a comparable expert-abstract corpus against which to compare the functional findings. Among retained hedges, no tokens were independently labeled politeness/deference (0 tokens). Softened recommendations were logged as false positives among recommendation frames (n = 15), so the null applies to claim-attached softening only. The zero count for politeness/deference is a result under a claim-restricted scheme, consistent with abstract genre compression, rather than a general finding about how these writers handle politeness or directives. Expert abstracts in the same three disciplines, coded with the same Step 1-3 scheme, would help separate genre effects from writer-competence effects.

This study analyzes the archived thesis abstract as a ratified textual product rather than a measure of individual cognition. Undergraduate theses are heavily supervised, and abstracts are often the product of multiple revisions guided by advisers. Consequently, observed hedging patterns do not solely reflect the independent rhetorical choices or internalized norms of developing researchers. They may also reflect the standards of the disciplinary community as enforced by faculty advisers. Therefore, while the study investigates patterns within texts produced by developing researchers, its claims are limited to the epistemic stance constructed in the final abstracts and the departmental standards they represent, rather than the internal cognitive processes of the writers themselves.

Definition of Terms

The following terms are defined conceptually and operationally to ensure consistency throughout the study:

Abstract, conceptually, is a concise, self-contained summary of a research work. Operationally, this refers to the complete English-language summary section of undergraduate theses archived in the PSAU institutional repository from 2023 to 2025. It contains research objectives, methods, results, and conclusions.

Adverbial Hedges, conceptually, are adverbs or adverbial expressions that qualify the scope, frequency, or certainty of a proposition. Operationally, these include *generally, typically, possibly, probably, relatively, and essentially*. These are coded in this study when they restrict the generalizability of a claim rather than describe a procedural norm.

Applied Life Sciences, conceptually, refer to disciplines that apply scientific knowledge to the study and management of living organisms and biological systems. Operationally, this refers to the three specific undergraduate programs at PSAU included in the study: Agriculture, Biology, and Veterinary Medicine.

Claim Responsibility, conceptually, refers to the epistemic obligation a writer accepts for the accuracy, truth, or limitations of a statement (Salager-Meyer, 1994). Writers use hedges to distribute this responsibility between themselves, their data, or external factors. Operationally, this serves as the interpretive lens for analyzing why specific hedges were chosen in the abstracts.

Clause-Level Hedges, conceptually, are syntactic structures that frame a proposition as tentative, approximate, or subject to qualification. Operationally, these include expressions such as it seems that, it is possible that, to some extent, and in our view. These are coded in this study when the entire clause functions to express the writer's epistemic reservation about the following or embedded proposition.

Corpus-Assisted Discourse Analysis, conceptually, uses systematic text collections to identify linguistic patterns, combining quantitative frequency measures with qualitative functional interpretation. Operationally, this refers to the manual frequency counting, functional coding, and cross-disciplinary comparison of hedging devices across 60 PSAU thesis abstracts using a Google Sheets coding template.

Disciplinary Variation, conceptually, refers to systematic differences in language use across academic fields arising from different epistemic norms, evidence types, and rhetorical conventions. Operationally, this refers to observable differences in hedging frequency, type, and function across Agriculture, Veterinary Medicine, and Biology abstracts in this corpus.

Epistemic Stance, conceptually, refers to the linguistic expression of a writer's degree of commitment to the truth, certainty, or reliability of a proposition. It signals the writer's judgment about the status of the knowledge they are presenting. Operationally, this refers to the aggregate pattern of hedging devices used in an abstract. A high frequency of hedges indicates a cautious or tentative stance, while a low frequency or absence of hedges records fewer overt markers of caution. It does not by itself establish absolute commitment, especially where findings are binary or measured.

Epistemic Weight, conceptually, is the strength of the writer's commitment to the truth of a proposition (how certain or tentative the claim is). Operationally, Step 2 first requires a claim: a statement about data, the world, or prior work that evidence could show to be wrong. Aims, directives, and routine procedures are excluded by kind. The Working Test then asks whether removing the device raises that claim's epistemic weight (epistemic caution). The Reader Test asks whether epistemic weight stays the same but the claim reaches the reader more bluntly or the writer takes sole ownership against prior work (politeness/deference). When a token both reduces commitment and softens interpersonal force, it is coded epistemic caution with a concurrent politeness/deference reading under the Working Test path, not as two independent test passes. Non-claims and tokens that meet neither check are logged as false positives.

Hedge Function, conceptually, refers to the rhetorical purpose a hedging device serves in its discourse context. Operationally, this study uses two function labels, adapted from Hyland's (1998) discussion of hedging functions (not a fixed published taxonomy): epistemic caution, which expresses reduced commitment to a claim's truth or accuracy, and politeness/deference, which shows interpersonal softening toward readers or prior research. False positives are residual rejects recorded when a candidate fails the claim condition or both removal checks.

Hedging, conceptually, is the use of a linguistic device that changes the writer's stance toward a proposition: either the writer's commitment to the truth of that proposition (Epistemic Caution) or the writer's relationship to readers or prior work with respect to it (Politeness/Deference).

Operationally, this study codes four grammatical types after Hyland (1998). These are lexical hedges, modal verbs, adverbial hedges, and clause-level hedges. A token is retained only when Step 2 shows such a stance change on a claim.

Lexical Hedges, conceptually, are verbs, adjectives, or nouns that introduce uncertainty or tentativeness at the propositional level. Operationally, these include verbs such as *suggest, indicate, appear*, and *seem*. They also include adjectives such as possible, probable, and likely, and nouns such as *possibility* and *assumption*. These are coded in this study when they express epistemic qualification of a claim rather than description or recommendation.

Modal Verbs, conceptually, are auxiliary verbs that express degrees of possibility, probability, or conditionality. Operationally, these include *may, might, could, would, should*, and *can*. These are coded in this study when they signal epistemic uncertainty about a claim rather than capability, obligation, or future procedure.

Normalized Frequency, conceptually, is a statistical method used to compare linguistic features across texts of different lengths by standardizing the count to a base unit. Operationally, this refers to the formula used in this study to compare hedge usage: Normalized Frequency = (Total Hedges / Total Word Count) x 100. For example, if an abstract contains 3 hedges and 258 words, the normalized frequency is (3/258) x 100 = 1.16 hedges per 100 words.

Operational Restatement (Threshold Restatement), conceptually, a phrase that only restates a result already fixed by the measure or a field-standard threshold, without a further uncertain inference. Operationally, evidential verbs used this way (especially indicating) are coded as false

positives rather than as lexical hedges. In the Coding Decision Rules this rule is also called the "indicating" tautology test.

Rescue-Hedging (Rescue Framing), conceptually, the use of hedges after an explicit non-significant result so that a treatment or finding is still presented as potentially beneficial. Operationally, the term is used in this study as a descriptive label for that pattern.

Softening, conceptually, is the reduction of force, directness, imposition, or interpersonal impact in a statement. Operationally, Step 2 tracks softening only on claims (statements about data, the world, or prior work that evidence could show to be wrong). Retained hedges are labeled epistemic caution and/or politeness/deference only, by the Working Test and Reader Test after the claim condition. Aims, directives, and routine procedures fail the claim condition by kind and are logged as false positives. The finding of no politeness/deference in this study refers to reader-oriented softening of claims, not to softened directives.

Token, conceptually, is a single occurrence of a linguistic form in a text, counted as an instance rather than as a distinct word type. Operationally, in this study a token is each coded instance of a hedge candidate recorded during analysis: a retained hedge after Stage 1 and Stage 2 checks, or a false positive that failed those checks. Totals such as 65 hedge tokens or 65 false-positive tokens refer to instance counts across the corpus, not to the number of different forms.

METHODOLOGY

This chapter describes the research design, institutional context, data collection process, and corpus selection for the study. Each methodological decision was justified by the nature of the research questions and the practical constraints of an undergraduate thesis.

Research Context

This study was situated at Pampanga State Agricultural University (PSAU), a state university specializing in agricultural and applied life sciences. Agriculture, Veterinary Medicine, and Biology are the core applied life science programs of the university, making them appropriate disciplines for an institutional corpus study.

The Philippine higher education context is relevant because English is the official medium of academic writing instruction at PSAU despite being a second language for most of the population. At PSAU, undergraduate writing instruction in the programs sampled for this study emphasizes structure and grammatical correctness more than stance. Hedging is not treated as a distinct rhetorical skill in the course materials reviewed for this project. Developing researchers writing thesis abstracts were therefore producing a genre that demanded hedging with limited explicit preparation for its epistemic requirements. These conditions made PSAU an appropriate institutional site for examining disciplinary hedging among developing researchers in a Philippine context.

The PSAU thesis guidelines available for this study prescribed standard formatting but did not set a rigid rhetorical template for abstracts.

Research Design

This study employed a descriptive mixed-methods design combining quantitative frequency analysis with qualitative functional interpretation (Biber et al., 1998). The design treated hedging as observable English language use in ratified abstracts. The quantitative layer identified hedging type and distribution patterns through frequency counts normalized per 100 words. This enabled cross-disciplinary comparison on a common scale. The qualitative layer involved functional coding using Hyland's (1998) framework to assign rhetorical purposes to the identified hedges. The interpretation of disciplinary patterns was then informed by two theoretical lenses. These are Salager-Meyer's (1994) concept of claim-responsibility and Hyland's (2005) model of stance and engagement. All coding was conducted manually using Google Sheets. This prioritized researcher judgment over automated processing for functional coding decisions, since automated tools cannot reliably assign hedge function from context, which was central to this study's research questions.

Research Process

The study proceeded through the following steps:

a. Requesting Permission. The researcher secured permission from the PSAU Librarian to access the physical archives of undergraduate theses. Due to institutional policies on intellectual property, soft copies of theses are restricted and cannot be provided electronically. Access is limited to on-site reading of

hard-bound copies. The researcher obtained specific authorization to photograph the abstract sections of the selected theses for research purposes.

b. Corpus Construction. Disproportionate stratified systematic sampling was used, with discipline as the stratification variable and fixed analytical targets per stratum to ensure equal representation regardless of population size. The target sample sizes were n=20 for Biology, n=20 for Veterinary Medicine, n=10 for Agriculture, Animal Science, and n=10 for Agriculture, Crop Science, totaling 60 abstracts. Within each stratum, a sampling interval k was calculated as k = N/n, where N is the stratum's target pool size (theses completed 2023-2025) and n is the required sample size. The intervals were as follows: Biology k=7 (145÷20=7.25), Veterinary Medicine k=5 (104÷20=5.2), Animal Science k=4 (38÷10=3.8), and Crop Science k=9 (94÷10=9.4). Sampling began from a randomly selected position around the middle of each stratum's catalog and proceeded forward by taking every kth thesis. Where the end of the stratum was reached before the required sample size, counting wrapped around to the beginning of the catalog and continued until the target sample was met. Disciplinary classification followed the degree program under which each thesis was officially submitted (e.g., B.S. Agriculture, B.S. Biology, D.V.M.), not a research topic, to ensure objective categorization.

c. Text Extraction. Abstracts were captured through on-site photography of the hard-bound thesis copies. These images were then converted to plain text using Optical Character Recognition (OCR) tools such as Google Drive or Adobe Scan. Each converted text was manually verified against the original photographs to correct transcription errors. Particular attention

was paid to verifying sentences containing potential hedge tokens, as OCR errors in these sentences would directly affect the coding results. Text files were anonymized using discipline codes (Acs1-Acs10, Aas1-Aas10, VM1-VM20, B1-B20) and organized by discipline.

d. Pilot Testing and Codebook Refinement. A pilot test was conducted on 4 abstracts not included in the final corpus to establish the overall coding scheme (see Appendix A). Decision rules for boundary cases were refined during main-corpus coding as specific ambiguities appeared and logged in the codebook. After the codebook was finalized, the full corpus was re-coded under the final rule set. Pilot abstracts were drawn from the PSAU library archives but from theses whose hard-bound submission year was before 2023, keeping them outside the study's inclusion window. The coded data were reviewed by the thesis adviser to verify consistency and accuracy with the decision rules. Discrepancies were resolved through discussion and consensus.

e. Data Analysis. Hedge identification, type classification, and functional coding were all conducted manually using a Google Sheets coding template. Each abstract was read in full before coding began to ensure hedges were interpreted in context rather than flagged as isolated tokens. Ambiguous cases were resolved using the predefined decision tree in Appendix A. Frequency counts were normalized per 100 words to enable comparison across abstracts of varying length. In a separate pass, each abstract was also coded for outcome type from its results language (Binary, Continuous, Mixed, Null/NS, Other), so that density patterns could be read against the type of finding reported. Those codes are recorded in the coding

spreadsheet and discussed in Part IV. While the quantitative analysis identified what hedges were used, the qualitative interpretation applied Salager-Meyer's (1994) claim-responsibility framework and Hyland's (2005) stance model to explain why these patterns occurred. Given the corpus size, findings were interpreted as indicative patterns within the PSAU context.

f. Timeline.

Phase	Duration
Permission and library access	1 week
Corpus construction and on-site photography	3 weeks
OCR verification and text preparation	2 weeks
Pilot testing and coding finalization	2 weeks
Full coding and frequency analysis	8 weeks (about 10 hours per week)
Writing and revision	8 weeks
Total	**About 5.5 months (24 weeks)**

Coding Decision Rules for Ambiguous Cases

Ambiguous forms may function either as epistemic hedges or as non-epistemic lexical items depending on context. Table 1 summarizes the decision rules applied to the main boundary constructions in this corpus.

Table 1

Decision rules for ambiguous hedge forms.

Token	Hedge When	Not a Hedge When
suggest / indicate	Expresses uncertainty about a finding or conclusion	Introduces a recommendation or action
suggest / indicate + need for	Says the evidence suggests a need may exist	Says what should be done or is needed
indicated by	Introduces a conclusion drawn from evidence	Points to evidence for a claim already stated
indicate / indicates that	Introduces an inferential conclusion drawn from analysis	Used as a synonym for showed/demonstrated (factive reporting)
indicating (participial)	Introduces an inference beyond the operational definition of the result	Restates the operational meaning of statistical significance
can / may	Expresses possibility	Expresses capability
generally / typically	Qualifies scope	Describes a procedural norm
appear / seem	Expresses result uncertainty	Indicates location

Research Corpus

This study used a text corpus rather than human participants. The corpus consisted of 60 undergraduate thesis abstracts from Agriculture, Veterinary Medicine, and Biology programs at PSAU. This study employed disproportionate stratified systematic sampling, with discipline as the stratification variable and fixed analytical targets per stratum (n=20 for Biology, n=20 for Veterinary Medicine, n=10 for Animal Science, and n=10 for Crop Science) to ensure equal representation regardless of population size. This meant the sampling rate varied by population size, prioritizing equal sample sizes over proportional representation. For the Agriculture discipline, the sample was equally split between Crop Science (n=10) and Animal Science (n=10) to ensure balanced representation of the sub-disciplines. Stratum intervals and the sampling procedure are given under Research Process, step b. For a manually coded functional analysis at a single institution, 60 texts allow sufficient depth of interpretation within the timeframe and resources of a BA thesis. The study treated findings as baseline descriptive data for institutional curriculum review.

Inclusion Criteria

• English-language undergraduate thesis abstracts

• Complete abstracts available in the PSAU library archives

• Clear disciplinary classification based on official degree program

• Hard-bound submission year (as recorded in the PSAU library catalog) between 2023 and 2025

Exclusion Criteria

• Non-English abstracts or abstracts with significant Filipino code-switching

• Graduate-level theses

• Theses submitted under interdisciplinary degree programs (for example, Environmental Science) where a single disciplinary affiliation cannot be determined

• Theses whose hard-bound submission year (library catalog) is before 2023 or after 2025

Ethical Considerations. Abstracts were accessed on-site at the PSAU Library with proper authorization. Student names were fully anonymized using discipline codes (Acs1-Acs10, Aas1-Aas10, VM1-VM20, B1-B20) before analysis began. Thesis titles that appear as quoted wording inside an abstract (source text) were redacted in the coded appendix where they identified the work. Residual title fragments, if any, do not include student names. The study analyzed linguistic patterns in text, not student performance or personal information. All data files were stored on password-protected devices and encrypted cloud storage accessible only to the researcher and thesis adviser. These files are retained for five years to support verification of coding decisions

by future researchers and to allow for follow-up comparative studies, after which they are permanently deleted in line with standard academic data retention practice.

RESULTS, ANALYSIS, AND DISCUSSION

This chapter presents the results, analysis, and discussion of the hedging patterns identified in the 60 undergraduate thesis abstracts across Agriculture, Veterinary Medicine, and Biology at Pampanga State Agricultural University. The patterns are read as evidence of how these writers used English to manage claim strength in a short, supervised genre. The chapter is organized into four parts, each corresponding to a research question. Part I presents the types of hedging devices found across the three disciplines. Part II examines how hedge frequency varies across Agriculture, Veterinary Medicine, and Biology. Part III identifies the rhetorical functions served by the hedging devices. Part IV discusses the main cross-disciplinary contrast in the ratified abstracts of developing researchers at PSAU.

Table 2

PSAU Thesis Repository: Population and Sampling Summary

Category	Biology	Veterinary Medicine	Animal Science	Crop Science	Total
Total repository	**275**	**206**	**131**	**216**	**828**

Category	Biology	Veterinary Medicine	Animal Science	Crop Science	Total
2023-2025 (target pool)	145	104	38	94	381
Pre-2023 (excluded pool)	130	102	93	122	447
Pilot sample (pre-2023)	1	1	1	1	4
Main sample	20	20	10	10	60

Table 2 presents the repository populations and the final sample distribution. The target pool for 2023-2025 totals 381 theses. Equal allocation gives each discipline a comparable sample size and keeps the smaller Animal Science population from being underrepresented. Pre-2023 theses (n=447) serve as the source pool for pilot abstracts to keep them distinct from the main study period.

Part I: The Formal Repertoire of Student Hedging

Across the 60 abstracts, retained hedges are few relative to the length of the texts, and the stance work they do is concentrated rather than spread

through every results line. The inventory is leaner than the abstract baseline Hu and Cao (2011) reported for English-medium applied-linguistics journals, though the fields and the writers are not the same. That comparison is only a rough external benchmark. It situates this undergraduate life-science sample as restrained in how often caution is marked in English, without claiming an equivalent genre or population.

Each retained token was classified into one of four formal types following Hyland's (1998) taxonomy: modal verbs, lexical hedges, adverbial hedges, and clause-level hedges. Table 3 presents the full type inventory for the corpus. For each type it lists the surface forms that appeared and the counts by discipline (Biology, Veterinary Medicine, and Agriculture).

Table 3

Hedge Types by Count and Discipline

Hedge type	Hedges	Biology	Veterinary Medicine	Agriculture
Modal verbs	can	0	0	5
	could	2	1	3
	may	3	0	3
	might	0	1	2
Lexical hedges	potential	10	4	5

Hedge type	Hedges	Biology	Veterinary Medicine	Agriculture
	suggest / suggests / suggesting	6	1	6
	implies / implying	1	0	2
	indicating / indicates	1	1	1
	likely	0	2	0
	inferring	1	0	0
	possible	1	0	0
Adverbial hedges	potentially	3	0	0
Clause-level hedges	—	0	0	0
Total		**28**	**10**	**27**

The repertoire is lexical first. Writers more often qualify claims with content words such as potential and the suggest family than with modal verbs, and almost never with adverbial hedges or clause-level packages. Modal use, when it occurs, is limited to a small set (can, could, may, might), and Agriculture supplies most of the can tokens. Table 3 holds the counts. The linguistic point is the narrow formal range: a few preferred forms do most of the caution work.

The three programs do not share the same mix within that narrow range. Agriculture balances modal and lexical caution more evenly, Biology leans heavily on lexical qualification, and Veterinary Medicine uses the same lexical preference but with far fewer occasions of either type. The cross-program difference at this level is not access to a wide hedge taxonomy. It is how often, and with which preferred form, writers mark caution at all.

The zero count for clause-level hedges fits the abstract as a compressed, impersonal reporting genre. Clause-level frames that put the writer's stance in the foreground (first-person forms such as I think or in our view, and more impersonal ones such as it seems that) are uncommon in science abstracts. The present texts favor short, result-centered sentences, and that register leaves little room for the longer clausal packages in which stance is often packaged in student essays or discussion sections.

The Scope and Limitations section already notes that these abstracts are ratified products shaped by adviser feedback. Against that background, the absence of clause-level hedges is better read as a feature of the genre and of local supervisory practice than as evidence that the writers lack the syntactic means to hedge at clause level. A similarly restricted range of devices has been reported in other developing-writer corpora (Hyland & Milton, 1997). In this sample, institutional and genre setting is the more proximate account of the formal profile in Table 3.

Part II: How Much Caution the Abstracts Display

Part II turns from the formal inventory to how much caution the abstracts actually display in English. The question is comparative across Agriculture, Veterinary Medicine, and Biology: how often retained hedges

appear, how often abstracts show none, and what else sits beside those patterns in the same inventory. Table 4 reports the counts. The paragraphs that follow read them as patterns of wording, not as ends in themselves.

Table 4

Hedge Frequency by Discipline

Measure	Agriculture	Biology	Veterinary Medicine	Total
Total abstracts	**20**	**20**	**20**	**60**
Words	4,300	4,509	5,060	13,869
Hedges	27	28	10	65
Hedges/100w	0.63	0.62	0.20	0.47
Zero-hedge abstracts	8	5	12	25
False positives	17	34	14	65
Unhedged claims	6	4	4	14

Biology and Agriculture mark caution with similar frequency overall, while Veterinary Medicine marks it far less often. In linguistic terms, Bio and Ag abstracts more often contain at least some epistemic softening of findings or implications, while many VM abstracts stay with direct results wording from start to finish. Table 4 records the densities. The practical reading is that the gap

between Agriculture and Biology is negligible in how the English behaves, and Veterinary Medicine is the low-caution corner of the sample.

Abstracts with no retained hedge make that contrast sharper. More than half of the Veterinary Medicine abstracts never open an epistemic softener under the coding rules, while Biology and Agriculture more often include at least one. The VM pattern is therefore not only “fewer tokens per text.” It is a higher share of abstracts whose English never leaves the unsoftened reporting of results.

Table 4 also records 14 unhedged claims: statements where caution was warranted but absent. Nine are generalizations beyond the data [G], and five are causal claims stated more strongly than the design alone supports [C]. Twelve of the 14 fall inside zero-hedge abstracts. Bare trial results and measured findings are not counted as unhedged claims. Routine results reporting does not by itself require caution under the coding scheme. Table 4.1 illustrates the two subtypes with excerpts from the coded corpus.

Table 4.1

Illustrative Unhedged Claims

ID	Discipline	Subtype	Excerpt	Why unhedged
B6	Biology	G	...making them a weak, unsuitable material to use in building walkways.	Ranks eco-bricks as unsuitable for a use beyond the measured drop-test result.

ID	Discipline	Subtype	Excerpt	Why unhedged
B9	Biology	C	…pure MSG and pure ethanolic leaf extract of Z. talanai caused the lesions / aberrations to the liver and kidney.	States causation more strongly than a single experimental design alone supports.
Acs10	Agriculture	G	…T3: NPK+14-14-14 was the most compelling treatment among all treatments…	Generalizes a ranking (“most compelling”) beyond a comparative description of the trial.

ID	Discipline	Subtype	Excerpt	Why unhedged
VM16	Veterinary Medicine	G	...Noni fruit ethanolic extract with 40% concentration has antibacterial property.	States the property as settled from the reported tests without modal or lexical softening.

In the zero-hedge abstracts, assertive reporting forms recur: showed, revealed, observed, and highest among them. These are reporting verbs and a superlative, not a closed booster checklist. They show how those abstracts often deliver findings in direct, unqualified wording. The same count was not run on abstracts that already contain hedges, so it describes the zero-hedge subset only and supports no comparison with hedged abstracts.

Caution is unevenly distributed across individual abstracts, especially in Agriculture. Most Ag texts show little or moderate softening, while Acs2 alone stacks several hedges after a non-significant result. When that one abstract is set aside, Agriculture no longer sits above Biology in how often caution appears. Any simple ranking that treats Agriculture as the most cautious program is therefore partly the effect of one densely hedged text rather than a stable program style.

Two further cautions apply. Abstracts within a program may share advisers, so patterns may reflect supervision as much as individual choice, and

spread within each program remains large. If potential is set aside, the remaining inventory shrinks and Veterinary Medicine's share of unsoftened abstracts rises further. Low rates of hedging are widely reported in undergraduate academic writing (Hyland & Milton, 1997; Qiu et al., 2024). In this PSAU sample the sharpest contrast in how English marks caution is between Biology and Veterinary Medicine. That local pattern appears in ratified abstracts, and supervised texts may reflect program tolerance for soft claims as much as individual rhetorical choice.

The pattern only partly matches prior cross-disciplinary work. Afshar et al. (2014) found agriculture using the fewest hedges among expert research articles. Here Agriculture is level with Biology in the full sample and looks more cautious only because of Acs2. Without that abstract it falls below Biology, which sits closer to Afshar et al.'s direction. Expert research articles and undergraduate abstracts are different genres, so the comparison stays loose. What the present table can support is a local reading of how three programs at one institution word caution, not a general claim about agricultural writing worldwide.

False positives also differ by program, and not in the same direction as retained hedges. Biology yields more rejected lookalikes, often potential in aim statements and indicated by as an evidential pointer: forms that resemble hedges in surface shape but do not mark uncertainty about a claim under the coding rules. A higher false-positive count therefore means more hedge-like vocabulary in non-hedging roles, not more hedging. Read with Table 4, Biology both softens claims more often when the token is retained and uses more

lookalike wording that the scheme rejects, while Veterinary Medicine is low on both retained softeners and rejected lookalikes.

Part III: What Hedging Does When It Appears

Function Profile of Retained Hedges

Part III asks what retained hedges do when they appear, and how the coding scheme separates them from lookalike forms. Table 5 reports how candidate tokens were classified after the claim condition and the two removal checks. Every retained token was coded as epistemic caution, and none was coded as politeness/deference. An equal number of candidates were logged as false positives because they failed the claim condition or both removal checks. On the retained side, the functional profile is uniform: when a token counts as a hedge in this corpus, it softens commitment to a claim about findings or implications, not interpersonal stance toward the reader.

Table 5

Function Distribution among Candidate Tokens

Function	Frequency	Note
Epistemic caution	65	Retained; passed Working Test
Politeness / deference	0	Null in this sample
False positive	65	Not retained (failed claim condition or removal checks)

Function	Frequency	Note
Total	**130**	**Logged candidate decisions**

That uniformity follows from the claim condition and the two removal checks, not from a claim that abstracts never soften directives in any linguistic sense. Only tokens attached to claims that pass the claim condition and at least one removal check enter the retained set. Recommendations and other non-claims are logged as false positives and never receive a politeness/deference label under this scheme. Fifteen recommendation-framed tokens sit in that false-positive pool. A wider definition of interpersonal softening could treat some of them differently. The all-epistemic-caution result fits how scientific English often uses hedges on knowledge claims (Hyland, 1996, 1998; Salager-Meyer, 1994): the devices manage how far a proposition should be trusted.

Gillaerts and Van de Velde (2010) show that research-article abstracts are not miniature full articles. Interactional markers in abstracts can be more limited than in the longer text. In this PSAU sample, retained hedges most often sit in implication and conclusion sentences rather than in bare results lines. Writers more often qualify what the findings mean for application or interpretation (potential, the suggest family) than they soften the reporting of the measured result itself. Results lines more often use factive reporting verbs without qualification. Table 5.1 shows how the same surface forms are kept or rejected once the claim condition and removal checks are applied. Table 5.2 then samples retained wording across the three programs.

Hedge versus False Positive in the Wording

Table 5.1

Retained Hedge versus False Positive: Coding Contrasts

ID	Token	Decision	Excerpt	Coding rationale
B1	potential	Retained (lexical)	…insights into the potential of B. subtilis as a sustainable solution…	Qualifies a finding-side claim about what the results support.
B1	potential	False positive (topic-naming)	In this study, the growth promoting potential of Bacillus subtilis… was assessed…	Names the object of study in an aim/methods frame (non-claim).

ID	Token	Decision	Excerpt	Coding rationale
Acs3	can	Retained (modal)	…it can be derived that the application of different alternative rooting hormones…	Possibility reading on an interpretive claim; removing can raises commitment.
Aas3	can	False positive (capability)	…rice bran mixed with water spinach… can help minimize…	Marks ability of a treatment, not uncertainty about a finding-claim.

ID	Token	Decision	Excerpt	Coding rationale
B3	indicating	Retained (lexical)	…presence of bacterial coliform and E. coli…, indicating unsafe conditions…	Inference beyond the operational lab result.
Aas8	indicating	False positive (restatement)	…lowest feed conversion ratio…, indicating superior feed efficiency…	Restates what the measure already means operationally.

The pairs in Table 5.1 are drawn from the coded corpus. In each case the surface form is similar. What changes is whether the host clause is a claim about findings or implications, and whether removing the token raises epistemic commitment. Topic-naming, capability, and operational restatement stay outside the retained set even when the word looks like a hedge.

Table 5.2

Illustrative Retained Hedges by Program

Discipline	ID	Token	Excerpt
Agriculture	Acs1	suggest	These findings suggest that fermented banana peel…
Agriculture	Acs2	could / potential	…the mycorrhiza treatment could still offer potential benefits…
Biology	B1	implies	This implies that the use of low concentration of B. subtilis…
Biology	B4	may	…flavonoids and saponins may contribute to its therapeutic effects.

Discipline	ID	Token	Excerpt
Veterinary Medicine	VM5	potential	...highlight the potential of M. calabura L. as a source of antifungal agents...
Veterinary Medicine	VM12	might	...might be effective... but it is not conclusive that ASF is not present...

Taken together, the English of these abstracts adjusts the epistemic weight of findings and implications when hedges are retained. It does not show retained hedges doing interpersonal softening of those claims. The stance repertoire that survives the claim condition and the removal checks is functionally narrow and centered on the proposition.

Part IV: Uneven Caution across Programs

Part IV asks what the density and function findings mean once outcome type and program wording are read together. The discussion draws on Salager-Meyer's (1994) claim-responsibility framework and Hyland's (2005) stance model. Table 6 classifies how each abstract reports its main results. Table 6.1 samples how the programs word caution in concrete excerpts.

Outcome Type and the Limits of a Design Explanation

Outcome type classifies how each abstract reports its main results from the abstract's own results language: binary presence/absence findings, continuous measures, mixed formats, null or non-significant results, and residual forms. Table 6 shows that outcome mix alone does not explain the program contrast. Veterinary Medicine is not simply a binary-only set, and Biology is not simply continuous-only. Many VM abstracts do report clinical findings from animal studies (drug-efficacy trials, disease detection, hematological profiling), and binary or presence-style results are common in that set, but they do not fully explain the wording gap. Biology abstracts with similar result styles still show more epistemic softening of implications. The thinner caution in VM is therefore consistent with departmental or supervisory practice, with outcome mix, or with both. It should not be read as a general rule that binary evidence needs no hedge under Salager-Meyer's (1994) claim-responsibility account.

Table 6

Outcome Type by Discipline

Outcome Type	Agriculture	Veterinary Medicine	Biology	Total
Binary	0	8	7	15
Continuous	4	4	2	10
Mixed	2	3	4	9
Null/NS	14	2	5	21
Other	0	3	2	5

Outcome Type	Agriculture	Veterinary Medicine	Biology	Total
Total	**20**	**20**	**20**	**60**

How Programs Word Caution

Where hedges do appear, they more often mark the move from result to implication than the bare reporting of a measured value. That tendency is easiest to see in Agriculture, where implication sentences frequently carry suggest, could, or potential. It is hardest to see in Veterinary Medicine. Many VM abstracts report findings in direct English and rarely open the kind of implication sentence where writers elsewhere add those softeners. That shows up in the Table 4 inventory as fewer retained hedges and more abstracts with none at all. Under Salager-Meyer's (1994) account, hedging an implication takes partial distance from a conclusion that goes beyond the measured finding. Thinner softening in the VM subset is therefore readable as less explicit negotiation of that responsibility in the ratified English of the abstract. It is not readable as more overclaiming: the unhedged-claim inventory is not higher for VM than for Biology (Table 4). Hyland's (2005) stance model fits the same linguistic picture: fewer hedges mean less explicit softening of authorial commitment in the text that remains. Whether the pattern comes mainly from supervision, from local reporting habits, or from both cannot be settled from the abstracts alone.

Reading abstracts with similar result styles side by side makes the contrast concrete. Biology excerpts more often carry an inferential step after the finding (implies, indicating where the inference exceeds the lab result). Veterinary Medicine excerpts more often stop at the finding or use a single

cautious modal in a limited frame. Acs2 shows the opposite pole inside Agriculture: after non-significance, residual benefits are still worded with stacked caution (suggests, might, could, potential), which keeps a softened positive reading in play after the statistical claim has failed. Table 6.1 places these wordings next to one another so the difference is visible in the English, not only in the program totals.

Table 6.1

Program Contrast: Sample Excerpts

Discipline	ID	Role in contrast	Token	Excerpt
Biology	B3	Retained hedge	indicating	…coliform and E. coli…, indicating unsafe conditions…
Biology	B1	Retained hedge	implies	This implies that the use of low concentration of B. subtilis…

Discipline	ID	Role in contrast	Token	Excerpt
Veterinary Medicine	VM2	Retained hedge	indicating	…92% … showed good attitude, indicating a strong sense of empathy…
Veterinary Medicine	VM12	Retained hedge	might	…might be effective… but it is not conclusive…
Veterinary Medicine	VM1	Zero-hedge abstract	—	No retained hedge token in the abstract under the claim condition and the removal checks.

Discipline	ID	Role in contrast	Token	Excerpt
Agriculture	Acs2	Stacked hedges	suggests / might / could / potential	…suggests that while statistical significance might not have been observed, the treatment could still offer potential benefits…

What the Contrast Can and Cannot Mean

Across the three programs, the contrast that stands out is therefore not a raw ranking of counts but a difference in how caution is worded around findings and implications. Biology more often sustains epistemic softening when results turn into meaning. Veterinary Medicine more often leaves that turn thinly marked or absent. Agriculture sits between those poles and occasionally stacks caution after a weak statistical claim. Abstract text alone cannot separate field convention, adviser practice, and individual skill as causes of that pattern.

Open Readings

Two readings follow as hypotheses for later work. First, VM abstracts may more often end on results with no implication sentence, which leaves less linguistic room to hedge. A move-structured count of the VM subset would test that. Second, advisers may filter interpersonal softening out of final abstracts, which would help explain why retained hedges in this corpus do only epistemic work. Comparing early drafts with the ratified abstracts would test that second reading.

CONCLUSIONS AND RECOMMENDATIONS

This study addressed a local gap: no published study identified in this review had mapped how developing researchers in Agriculture, Veterinary Medicine, and Biology at PSAU manage uncertainty in English thesis abstracts. From that gap, the inquiry traced what types of hedging appear in those abstracts, how often they occur across the three programs, what functions the retained devices serve, and what those patterns suggest about epistemic stance in the ratified text. The analysis drew on Hyland's (1998) formal types, treated Epistemic Caution and Politeness/Deference as the two forms of stance change toward a proposition, and used Crompton's (1997) restatement logic to inform the Working Test. Sixty complete English abstracts, twenty per discipline, were coded through a wide net of candidates, a claim condition with paired removal checks, and type assignment on retained hedges only.

Conclusions

This study set out to map how developing researchers at PSAU encode caution in English in the thesis abstract, a short genre where every claim is visible and already ratified by advisers. The picture that emerges is a constrained practice. Writers often state results with little or no softening. When they do hedge, they reach for a small lexical set and use it mainly to qualify what the findings might mean, not to elaborate stance at length. Under the coding rules used here, that work is epistemic. In these abstracts, writers report findings and edge implications.

Disciplinary difference exists, but it is thinner than a strong “three cultures” story would predict. Veterinary Medicine is the clearest low-caution corner of the sample: its abstracts more often leave findings and implications unsoftened. Agriculture and Biology are less distinct from each other once Acs2 is read as one densely hedged text rather than as proof of a whole-program style. The deeper point is institutional as much as linguistic. These texts are supervised products, so what looks like a writer’s caution profile may also be a program’s tolerance for soft claims in a high-stakes summary.

The contribution is therefore a baseline, not a verdict on competence or on whole disciplines. For BAELS and related instruction, the finding matters because it shows what "hedging in science abstracts" looks like on the ground at this campus: uneven, lexically narrow, and easy to miss if teaching assumes a full academic repertoire. For research, the same baseline makes later comparison possible, against expert abstracts, other programs, or fuller thesis chapters where other functions of mitigation may have more room.

Recommendations

Based on the findings of this study, writing instructors and thesis advisers at PSAU may treat hedging as a judgment about how far a claim may go beyond the data, rather than as a general polish added for caution alone. Feedback on abstracts may focus on the move from results to implication. That is where this corpus shows both missing caution on generalizations and causal claims, and denser hedging that keeps non-significant treatments sounding still useful. Advisers already shape the final abstract. Making that judgment explicit in supervision may help students see when a finding can stand plain and when an implication needs restraint.

Teachers in the BAELS program and related writing courses may use the local pattern as a teaching resource. Many abstracts in the sample soften little. When they do, the work is concentrated in a narrow set of forms, mainly lexical items such as *potential* and *suggest*, with modal verbs as the other main resource. Instruction that assumes a full academic hedging repertoire will miss what this campus sample shows. Classroom tasks may instead train students to match the strength of a claim to the strength of the evidence in the same sentence.

Future researchers may use this study as a local baseline for undergraduate thesis abstracts in Agriculture, Veterinary Medicine, and Biology at PSAU. They may expand the design with inter-rater coding on at least a subset of the corpus, with independent proficiency measures, or with a matched set of expert abstracts from the same three programs. Those steps would help separate genre compression in the abstract from writer resource limits and from adviser-enforced departmental standards.

Researchers may also apply the coding rules and false-positive log from this thesis to longer genres, such as discussion chapters, where interpersonal softening of claims may have more room than the abstract form allows.

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